

RAZ EINI

Software Engineering Student • Security, Systems & Backend Engineering

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PROFILE

Software Engineering student (Atuda) targeting cyber and intelligence units, with hands-on experience in defensive security, low-level performance optimization, and cryptographic implementations. Demonstrated ability to build production-grade tools, including a hybrid Python/C NIDS with native C acceleration (8x speedup, memory-safe) and Grafana observability, alongside a cryptographically secure CLI vault with automated CI/CD pipelines. Actively researches binary exploitation and reverse engineering through picoCTF challenges. Proficient in Linux environments and driven to apply low-level security concepts to high-impact cyber operations.

SECURITY & TECHNICAL SKILLS

- **Security & Observability:** Deep Packet Inspection (DPI), Native C Acceleration, NIDS/IPS logic, Active Threat Mitigation (Dynamic IP Isolation), Shannon Entropy Analysis, Bitwise TCP Flag Analysis, Grafana & Loki (Dashboards as Code), Promtail.
- **Binary Exploitation & Reverse Engineering:** Buffer Overflows, Control Flow Hijacking, Shellcoding, Memory Corruption, Dynamic Debugging (GDB + GEF), Disassembly (Ghidra, objdump), Exploit Automation (pwntools).
- **Networking & Protocols:** Packet Capture & Analysis (Scapy, Raw Sockets), OSI/TCP-IP Stack, Dynamic Memory Garbage Collection, Thread-Safe Concurrency (threading, queue.Queue).
- **Cryptography & Systems:** Fernet (AES-128-CBC & HMAC), PBKDF2-HMAC-SHA256, Salting & Hashing, Key Derivation Functions (KDF), Linux Systems & CLI Administration, Binary Analysis, PyPI Package Publishing.
- **Languages & Infrastructure:** C, Python, Java, Kotlin, Bash, ctypes, Git & GitHub Actions (CI/CD), Docker & Docker Compose, Android (Jetpack Compose, MVVM), Firebase.

PROJECTS

NetGuard – Hybrid Python/C NIDS & Observability Engine [View on GitHub ↗](#)

Python, Native C (ctypes), Scapy, Docker Compose, Grafana, Loki

- Architected a high-performance Network Intrusion Detection System (NIDS) with dynamic active defense (IP isolation) and multi-threaded packet processing.
- Engineered a bounds-checked native C DPI engine via ctypes (batch processing) to bypass Python FFI overhead and GIL bottlenecks, achieving an 8x throughput speedup (~2.5M pkts/sec) on raw traffic.
- Implemented **Shannon Entropy math** to detect DNS Tunneling/covert channels, alongside bitwise TCP flag analysis for stealth scan detection (NULL, FIN, XMAS).
- Designed a producer-consumer pipeline (queue.Queue, threading.Lock) with a zero-leak background garbage collector, shipping structured JSON logs to an automated **Grafana/Loki** observability stack.

CTF Writeups – Binary Exploitation & Reverse Engineering [View on GitHub ↗](#)

C, Python, pwntools, GDB (GEF), Ghidra, Linux (WSL2/Ubuntu), Bash

- Maintain a structured, modular repository of picoCTF writeups covering assembly & GDB fundamentals, memory corruption, and reverse engineering.
- Analyze unknown compiled ELF binaries to identify memory-corruption flaws (buffer overflows, format-string bugs) and craft reliable exploits to hijack control flow.
- Combine dynamic analysis (GDB + GEF) with static analysis (Ghidra, objdump) to reconstruct runtime stack state and automate exploit delivery with pwntools.
- Document each challenge with checksec output, stack-layout diagrams, and local/remote-capable exploit scripts, following a consistent, professional writeup format.

Secure CLI/TUI Password Manager (PyPI Package) [View on GitHub ↗](#)

Python, Cryptography, CI/CD, PyPI

- Architected a modular password manager using a clean src-layout structure, ensuring robust separation between the cryptographic core and CLI interfaces.
- Implemented high-security standards including PBKDF2-HMAC-SHA256 key derivation (390k iterations) and Fernet (AES-128-CBC + HMAC) authenticated encryption to ensure data integrity and confidentiality.
- Automated end-to-end CI/CD release pipelines via GitHub Actions to run pytest suites and publish production-ready packages directly to PyPI.

E-Commerce Shop App [View on GitHub ↗](#)

Android, Java, Firebase

- Developed a scalable shopping application with real-time Firebase database synchronization, multi-device cart consistency, and role-based admin inventory management.
- Implemented Firebase Authentication and a responsive Java UI layer (RecyclerView, Fragments) to support a smooth end-to-end checkout flow.

Smart TaskFlow [View on GitHub ↗](#)

Kotlin, Jetpack Compose, MVVM

- Engineered a task management application using Jetpack Compose and MVVM architecture, with automated prioritization logic and clean, decoupled business logic.
- Structured state management with reactive ViewModels and StateFlow, keeping UI recomposition efficient and business logic fully testable.

EDUCATION

B.Sc. Software Engineering, Afeka College of Engineering | Present | Expected Graduation: 2029

Academic Reserve (Atuda) "Amirim" Program — pre-service technological track

Makif Tet High School | 2022–2025

- High School Diploma (GPA: 91): Majors in Software Engineering & Cyber (10 units), Physics, Math, and English (5 units).
- Key Achievements: Grade 100 in Systems Design & Programming; Software Technician Certification.

ADDITIONAL

- **Languages:** Hebrew (native), English (high level)
- **Interests:** Chess (Peak 2200 ELO), Speedcubing